

Unified Military Analytics and Comparison Dashboard-DV

Report for Milestone 4

**INTERNSHIP OF
DATA VISUALIZATION
IN
INFOSYS SPRINGBOARD**

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ABSTRACT

This project aims to develop a fully interactive dashboard suite for analyzing global military power in 2025. Using Python, military metrics will be scraped from GlobalFirepower.com for 140+ countries via a pre-defined list (links_for_military_data.txt).

The project leverages Python-based data pipelines to scrape, clean, and process open-source defense data from GlobalFirepower.com. Unlike traditional single-platform dashboards, GDM emphasizes cross-platform flexibility — allowing deployment in tools like Tableau, Power BI, or Python-based web apps built using Streamlit or Dash.

Dashboards planned:

- Quick Stats – A global overview of rankings, trends, and highlights.
- Nation Overview – Detailed analysis of individual country capabilities.
- Compare Powers – Side-by-side military and economic comparison.
- Coalition Builder – Interactive simulation of alliance strength and combined assets.

The final deliverable is a comprehensive analytics suite that enables users to compare nations, evaluate coalitions, and analyze KPIs like Power Index Rank Gap, Assets per Capita, and Defense Budget-to-GDP Ratio through intuitive and dynamic visualizations.

Milestone 4: Final Review and Delivery

Module 7: Testing and Debugging

QA checklist:

i. Quick Stats Dashboard:

- Filters Testing
 - Region filter updates all charts correctly
 - Continent filter works properly
 - Alliance (NATO) filter shows correct countries
 - Multiple filters together work without errors
- KPI Validation
 - Total Defense Budget matches dataset
 - Total Military Manpower is correctly summed
 - Budget to GDP Ratio calculated correctly
 - Assets per Capita calculated correctly
 - Country Count changes based on filters
- Charts & Visuals
 - Top 5 Active Personnel pie chart shows correct percentages
 - Top 10 Power Index line chart sorts correctly
 - Defense Budget pie chart shows correct values
 - Assets per Capita chart displays correct countries
 - Map bubbles match manpower values
- Layout & UI
 - Titles are clear and correctly spelled
 - Axis labels are readable
 - No overlapping text
 - Tooltips show correct units (USD, bn, %, etc.)

ii. Nation Overview Dashboard:

- Filters Testing
 - Country dropdown updates all visuals
 - Alliance filter works correctly
 - Map highlights selected country
- KPI Validation
 - Total Land Area matches dataset
 - Total Population matches dataset
 - Power Rank matches source data
 - Budget per Soldier correct
 - Budget to GDP Ratio correct
- Charts & Visuals
 - Active vs Reserve vs Paramilitary pie chart correct
 - Military Participation Ratio gauge shows correct value
 - Defense Budget bar chart accurate
 - Naval, Tank, Aircraft chart accurate
- Layout & UI
 - Titles are clear and correctly spelled
 - Axis labels are readable
 - No overlapping text
 - Tooltips show correct units (USD, bn, %, etc.)

iii. Compare Powers Dashboard:

- Filter & Parameter Testing
 - Country A selector works
 - Country B selector works
 - Both countries can be changed independently
 - Charts update instantly when country changes
- KPI Validation
 - Manpower for Country A correct
 - Manpower for Country B correct
 - Assets per Capita correct for both
 - Budget to GDP ratio correct
 - Rank gap calculated correctly

- Charts & Visuals
 - Defense Budget charts show correct values
 - Active personnel charts accurate
 - Aircraft, Tank, Naval fleet comparison correct
 - Colors clearly differentiate Country A and B
- Layout & UI
 - Side-by-side comparison aligned properly
 - No overlapping bars
 - Labels readable

iv. **Coalition Builder Dashboard:**

- Filters Testing
 - Coalition A multi-select works
 - Coalition B multi-select works
 - Countries can be added/removed easily
 - Aggregated totals update correctly
- KPI Validation
 - Total Manpower = sum of selected countries
 - Total Land Area = correct sum
 - Total Naval Fleet = correct sum
 - Total Aircraft = correct sum
 - Total Tanks = correct sum
 - Total Defense Budget = correct sum
- Charts & Visuals
 - Coalition A and B shown separately
 - Bars represent total values correctly
 - No duplicate counting
 - Units clearly displayed
- Layout & UI
 - Coalition A and B clearly labelled
 - No visual clutter
 - Consistent color scheme
 - Dashboard loads without lag

Module 8: Documentation and GitHub Release

There are total 4 dashboards. Each dashboard can be accessed by ctrl+click on the navigation buttons.

i. Quick Stats

Purpose: Provides an overall summary of global military data.

How to Use:

- Use the filters at the top (Region, Continent, Alliance).
- KPI cards update automatically based on selected filters.
- Hover over charts to see detailed values.

ii. Nation Overview

Purpose: Shows detailed analysis for a selected country.

How to Use:

- Select a country from the Country dropdown.
- All KPIs and visuals update automatically.
- View:
 - Total land area
 - Population
 - Power rank
 - Budget per soldier
 - Military participation ratio
- Charts display personnel distribution, defense budget, and equipment data.

iii. Compare Powers

Purpose: Compare two countries side by side.

How to Use:

- Select Country A.
- Select Country B.
- The dashboard compares:

- Military manpower
- Defense budget
- Assets per capita
- Tanks, aircraft, and naval fleet
- Power rank gap
- Charts update instantly when selections change.

iv. Coalition Builder

Purpose: Compare two groups of countries.

How to Use:

- Select multiple countries for Coalition A.
- Select multiple countries for Coalition B.
- The dashboard calculates total:
 - Manpower
 - Land area
 - Tanks
 - Aircraft
 - Naval fleet
 - Defense budget
- All totals are automatically aggregated.

Finally published to Power BI online

